

Business Metrics Discovery by Business Rules

Francesco Arigliano³, Paolo Ceravolo¹, Cristiano Fugazza¹, and Davide Storelli²

¹ Università degli studi di Milano, Dipartimento di Tecnologie dell'Informazione
via Bramante, 65
26013 Crema (CR), Italy
{ceravolo, fugazza}@dti.unimi.it

² e-Business Management Section of S.S. ISUFI, University of Salento
via per Monteroni, s.n.
Lecce, 73100, Italy
davide.storelli@ebms.unile.it

³ Engineering Ingegneria Informatica S.p.A., R&D Lab,
via Trinchese, 61
Lecce, 73100, Italy
francesco.arigliano@eng.it

Abstract. This work contributes to the results of the TEKNE projec¹, a project aimed at developing a framework for Business Process Management (BPM), supporting the designer with a set of *performance indicators*. The indicators drive the designer in estimating if the process comply to the objectives and when necessary enable re-engineering of the process. In particular this paper discusses how to derive performance indicators directly from requirements expressed in a Business Rules (BR) format.

1 Introduction

In modern organizations business process (BP) design and management is more and more a critical asset. Global competition stresses to the adoption of continuing design methodologies, where business processes are rapidly re-engineered to conform to the evolution of marked demand. In this context it is straightforward to see the relevance of business process monitoring.

In principle monitoring can be done in many ways; traditional approaches rely on interviews to the actors executing the process or on the analysis of market data. But nowadays much interest is on using informative systems (IS) to support business process management. Among the benefits provided by such an approach is a strong support in monitoring the business execution. Indeed, all the events generated in the enactment of a process are recorder in a log. This provides a continuing feedback on the actual development of the process that can be compared with requirements such as the *BP model* or *directives* (goals, strategies, standards, and regulations) that the process need to comply with. Note that the information obtained is complementary to the one in the business process model because, at the design time, one may model

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optional and alternative activities. Also, some requirements cannot be enforced directly in the process design because of their external origin (standards, regulations, protocols, etc), or because of a lack of control on activities (outsourcing, user decision, etc). In short, the event logs can radically improve the effectiveness of monitoring tasks, reducing the time of analysis and facilitating re-engineering of the process.

But drawbacks exist. The main problem is how to relate the events generated by the execution layer to business activities in the process model. In fact, log data can provide insightful information only if a univocal relation is established between events and activities. In traditional approaches, event logs and business activities are completely uncoupled. However, in the literature, several works propose the integration of BPM with an execution layer. In general we can distinguish two approaches. The first is based on the adoption of procedural languages aimed at describing business process policies. These approaches rely on a formal description of the business process and support the monitoring of formal properties, such as for instance, termination, cycles, and transaction. The formal representation guarantees algorithms and techniques for driving the analysis but requires an ad hoc mapping among formal policies and the business requirements to be monitored.

The other approach is based on enriching the event log with metadata relating them to business activities. This way, we have a closer connection between requirements and event logs. But the adoption of metadata introduces an additional layer. Indeed in a business context requirements are represented using business rules (BR) standards, traditionally implemented with natural language eventually coupled with a graphical notations, while events logs refer to business activities via annotations, in form of tags or simple assertions.

Our approach is aimed at reducing the gap between the representation of event logs and BR. The platform we developed supports the design of BR, BP and IS in a single framework, where the relation among all these notations is explicit.

In particular, our approach identifies a specialization of the BR, named *directives*, aimed at describing constraints to be applied to the BP in order to comply with a given objective, strategy, or regulation. Directives are used to derive performance indicators aimed at assisting the designer with measures on the process, to be used in the re-engineering phase.

The paper is structured as follows: the first section introduces performance indicators discussing the methodology adopted in the TEKNE project; the second section details the architecture; while the third presents the related works.

2 Monitoring the Process

2.1 Performance Indicators

As proposed in [1], process monitoring techniques can support three services: *discovery*, *conformance* and *extension*. *Discovery* is about deriving a process model from data in the event log. Using an inductive approach, the process executions are generalized in a model consistent with all the recorded instances. *Conformance* is about verifying if the executions follow the prescribed control flow or comply with external directives such as business goals, regulations, standards etc. In this case, the information gathered by

logs must be compared to the business model and to the business rules; moreover, when a violation is detected it is pointed out. *Extension* is about enriching the business model and the business rules with new constraints derived from the process instances described in logs.

This work develops a *conformance* approach, the problem of evaluating performances can be described by the following implication:

$$S, W \rightarrow R \quad (1)$$

Where S is a specification of the behavior that must result from execution of the process to be developed; W represents a set of world properties, i.e. the context where the process is executed; and R represents a set of requirements to be met by the process. To prove that the specifications will satisfy the requirements in a specific world context, it is necessary to show that this implication holds.

As described in details in following sections our framework provides the designer with support for the specification of all these descriptions. In particular the BXModeler editor allows to describe the BP model relating it with an execution layer (IS). This editor is coupled with XBeaver, an editor for vocabulary and BR allowing to describe both the business context (W) and the requirements of the business (R).

The aim of our system is to verify that the implication 1 holds. This can be done both *ex-ante* and *ex-post*. *Ex-ante* is acting on the process specification only, for instance by constraining the naming of objects in conformance with the entities declared in the business context, or verifying that the overall set of BR is consistent. These indicators are implemented without considering the execution phase; for this reason they lay outside the scope of this paper. Instead we concentrate on conformance indicators that are obtained by comparing the process specification to the process execution, i.e. *ex-post*. Using such an approach the process S to be tested is not the BP model. Indeed S represents an instance of BP execution. Moreover R can represent both the BP model and the directives. In the first case we evaluate if the execution conforms to the design, in the second case we evaluate if the execution conforms to a constraint not contained in the BP model.

A typical method adopted in order to implement conformance indicators is to identify a set of requirements and verify which executed instances of the process violate it. The notion of *violation* is very useful because it provides a simple criterion of measurement. Given a set of requirements R if the Implication (1) does not hold, S and W violate R . Note that this notion has an important property. Pointing out violations, we obtain a measure of the process performance with a method that is completely independent from the specific semantics of R , so we can support as many indicators as the number of requirements sets R we are able to represent. In other words, our methodology is independent of the specific requirements to be evaluated.

A methodology independent from requirements is crucial for our goals, because, as said, one of the aims of the TEKNE project is to directly use the BR to derive performance indicators. But such a methodology is not the only condition to be supported in order to use BR directives directly as performance indicators. The other condition is providing an algorithm, which is able to conjunctively evaluate S , W , and R , as a single knowledge based. This algorithm has to work as a black box taking in input S , W , R , computing the consistency of the knowledge based, and in case of inconsistency point

out a violation. Such an algorithm is compatible only with a representation of S , W and R in declarative form, and assuming a uniform naming space.

The algorithm verifying violations is executed by a theorem prover representing the knowledge base S , W , R , as a set of logical assertions. This approach is compatible with the standards adopted.

BXModeller represents the process in Business Process Modeling Notation (BPMN²), this notation is based on a flowcharting technique that is tailored for the purpose of creating graphical models of business operations. A straightforward way for translating it in logical assertions is to represent it using a logic-based formalism (specifically, OWL DL data structures). For this purpose, we shall use the Business Management Ontology (BMO³) as a means to express the structural properties of the process. XBeaver directly adopts a declarative approach because the BR language implemented by the tool (SBVR) was natively designed to be represented in First Order Logic.

2.2 Implementing Performance Indicators

At a first glance the notion of violation could appear poor, because related to the evaluation of a Boolean condition. But considering the occurrence of a specific violation it is possible to develop more complex indicators. In [4] some of us discussed the adoption of an approach based on violation for evaluating alethic and deontic conditions with either relative or absolute thresholds. Alethic directives are used to model necessities that cannot be violated, even in principle. For example, an alethic directive may state that an employee must be born on at most one date. Deontic directives are used to model obligations (e.g., resulting from company policy) which ought to be obeyed, but may be violated in real world scenarios. For example, a deontic directive may state that it is forbidden that any person smokes inside any company building. In order to support indicators expressing both alethic and deontic directives two behaviors are to be implemented. The primer must spotlight non-conformity of the process any time a violation is detected. The former must report on the occurrence of a violation. A directive expressing an absolute threshold prescribes the exact occurrence of a given event; while a directive expressing a relative threshold prescribe that the occurrence of a given event must be in a relation (i.e. a percentage) with another event.

Unfortunately representing this category of modal constraints in logical format suitable for a theorem prover it is not straightforward. Indeed it is well acknowledged that the introduction of modalities and disequations in a logical language lead to hard computational complexities.

In order to face this problem we dissociate the representation of constraints from the logical assertions representing directives. In other words, starting from a directive expressed as a BR we derive a twofold representation: (i) a logical assertion in universal form, to be evaluated by the theorem prover, plus (ii) a constraint representing the suitable modality or threshold to be supported. In particular we generate a *constraint function* f of the cardinalities of the facts generated by the process flow to be

² BPMN web-site: <http://www.bpmn.org>

³ Jenz and Partner. Business Management Ontology (BMO). Technical report, 2004.

used as an indicator of the process performance. Then, this cardinality is linked to thresholds expressed by directives. Equation (2) formalizes this point as follows:

$$f(Card_{A_1}, Card_{A_2}, \dots, Card_{A_n}) \leq \alpha. \quad (2)$$

This equation is not evaluated directly by the theorem prover but in a module dedicated to constraint functions.

Considering for example a business rule like the following:

The number of customers that apply for a Premium Card **must** be at least 90% of the number of customers that **do not have** a Premium Card. (3)

It can be represented by a universal assertion:

all customers apply for a Premium Card; (4)

plus a constraint in form of disequation:

customers that apply for a Premium Card
 $\geq 0.9 * \text{customers that do not have a PremiumCard.}$ (5)

In particular, Figure 1 shows how the system interrelates the theorem prover with the module tasked to represent constraints and the module tasked to evaluate them. Basically given a specific execution of the process S_j , the system starts evaluating all the directive violated by a knowledge base composed by S_j plus the business context

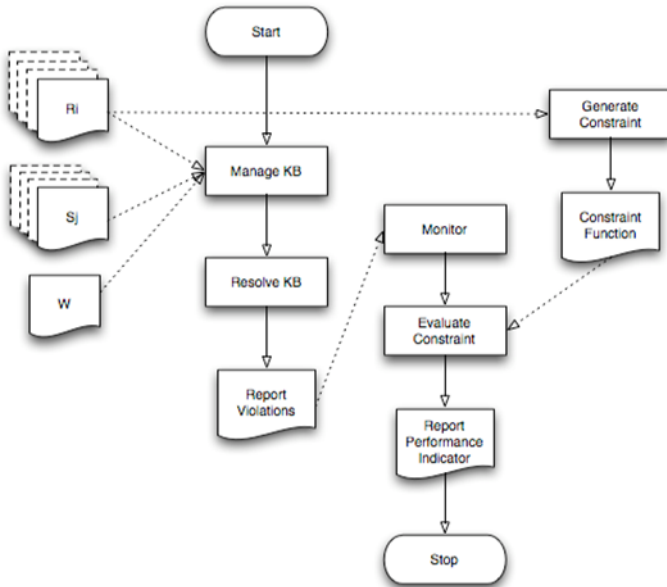


Fig. 1. The module generating performance indicators

W. In parallel a directive is translated in a constraint of the form described by Equation (2). When all instances of the execution of process S are evaluated by the prover, the system compares for each directive the occurrences of violations with the constraint and generate an indicator on the compliance of the process execution.

3 The TEKNE Architecture

The proposed architecture is compound by three parts that address the specification, execution and monitoring of a process, of its business context and of the expected requirements. The first part supports business engineers on design-time. It provides components and tools for modeling any business process together with its context and directives. The control flow is defined through the BXModeller, a collaborative web-application that represents processes through OMG BPMN (Business Process Modeling Notation) and exports their serialisation to a standard format for workflow execution: the XPDL 2.0 (XML Process Definition Language). The business context and the directives are described through the XBeaver application, that adopts a declarative approach, leveraging SBVR (Semantics of Business Vocabulary and Business Rules), another OMG standard. The adoption of SBVR is motivated by two features: i) SBVR models can be expressed by means of an easy to use natural-language-based notation; ii) the SBVR metamodel is compliant and mapped to first order logic, thus fully supporting automatic interpretation and reasoning upon its assertions. The combination of both features allows the definition of automatable metrics that can be referred directly to business policies and business rules. The architecture provides, indeed, a tool for coupling elements of a business domain with activities of a business process, thus binding generic logical assertions to generic operations. The second part of the architecture is constituted by the run-time environment in which the XPDL representation of the process is executed and the workflow operations are recorded in a log repository. This allows making important information available for analysis and metrics measurement. Thanks to the previous mapping phase among elements of the BPMN and of the SBVR models, it is possible to derive rules from actual instances of process activities. The resolution of rules is of responsibility of the third major part of the proposed architecture: the Metrics Framework.

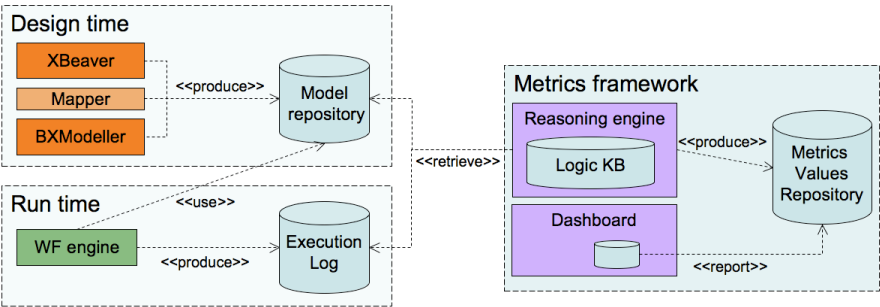


Fig. 2. A synthetic view on the proposed architecture

Here data from the design-time and run-time repositories are retrieved, merged and transformed into a logical formalism, thus constituting a knowledge base where logical assertions can be evaluated and metrics can be measured. Moreover a dashboard is provided in order to allow business managers to directly monitor the actual values of the metrics they have previously defined.

4 Related Works

Approaches aimed at supporting business process monitoring by an execution layer are not new in the literature. For example a wide literature exist on workflow mining [1]. In the web services literature there are several approaches dealing with monitoring the communication over service enabled business processes [8][9], verifying the correct execution of the prescribed flow, specifying coordination with other applications or internal time of response [7][10][13]. Advanced researches proposed dedicated policy languages for describing conditions.

Other works have focused on the introduction of a metadata layer supporting an enriched description of processes. Various authors [3][12] use planners over service description in DAML-S. In [5] a complete architecture for the analysis, prediction, monitoring, control and optimization of process executions in BPM Systems is presented. In [6] [11][2] we have one of the first proposals and implementation in the direction of integration Semantic Web technologies and BP.

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